Exercises for "Identification in Macroeconomics" Course

Fall 2026

Question 1

The main point of this question is to get you to familiarize yourself with perfect-foresight sequence-space techniques. This is not a course about HANK models, so you will instead be asked to code up simple representative-agent models. The application is a canonical one: monetary policy transmission to household consumption.

We consider the classical textbook small-scale New Keynesian model, but with a potentially non-standard household structure. The supply side of the economy is summarized by a standard Phillips curve:

$$\widehat{\pi}_t = \kappa \widehat{y}_t + \frac{1}{1+\bar{r}} \widehat{\pi}_{t+1}, \tag{1}$$

where hats denote log deviations from steady state, bars indicate the model's deterministic steady state, and $\bar{r}$ denotes the steady state real interest rate. Nominal interest rates are set according to a standard Taylor rule with interest rate smoothing:

$$\widehat{i}_t = \rho_i \widehat{i}_{t-1} + (1-\rho_i)\left(\phi_\pi \widehat{\pi}_t + \varepsilon_t^m\right), \quad \varepsilon_t^m \sim N(0, \sigma_m^2). \tag{2}$$

Here ε_t^m is a monetary policy shock. The government issues nominally risk-free bonds and collects lump-sum tax revenue. Written in real terms, the government budget constraint is

$$b_t + \tau_t = \frac{1+i_{t-1}}{1+\pi_t} b_{t-1}, \tag{3}$$

where b_t denotes real government debt and τ_t is tax revenue. Log-linearizing:[1]

$$\bar{b}\widehat{b}_t + \bar{\tau}\widehat{\tau}_t = (1+\bar{r})\bar{b}(\widehat{b}_{t-1} + \widehat{i}_{t-1} - \widehat{\pi}_t). \tag{4}$$

Unless indicated otherwise we will assume that the government always keeps government debt at steady state, so $\widehat{b}_t = 0$ for all t. Lump-sum taxes $\widehat{\tau}_t$ then adjust residually so that the government budget constraint holds.

Your job will be to characterize the dynamic (perfect-foresight) behavior of the economy in response to a one-off, date-0 monetary policy shock, $\varepsilon_0^m = -1$. You will do so for several variants of the model's (as-of-yet unspecified) household block. For all of those variants, the

[1] As in the usual textbook case we will consider a steady state with zero inflation, so $\bar{r} = \bar{i}$ and $\bar{\pi} = 0$.

household budget constraint will be

$$c_t + b_t = y_t - \tau_t + \frac{1 + i_{t-1}}{1 + \pi_t} b_{t-1}. \tag{5}$$

Log-linearizing:

$$\bar{c}\widehat{c}_t + \bar{b}\widehat{b}_t = \bar{y}\widehat{y}_t - \bar{\tau}\widehat{\tau}_t + (1 + \bar{r})\bar{b}(\widehat{b}_{t-1} + \widehat{i}_{t-1} - \widehat{\pi}_t). \tag{6}$$

For the remainder of this exercise you should assume the following parameter values: $\kappa = 0.05$, $\bar{r} = 0.01$, $\rho_i = 0.8$, $\phi_\pi = 1.5$, $\bar{c} = \bar{y} = \bar{b} = 1$, $\bar{\tau} = \bar{r} \times \bar{b}$, $\sigma_m = 1$.

a) Begin with the standard representative-household, permanent-income model. That is, the representative household has preferences

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma} - 1}{1 - \gamma}. \tag{7}$$

Note that β in this case is given as $\beta = \frac{1}{1+\bar{r}}$. Given sequences of income $\boldsymbol{y}$, nominal rates $\boldsymbol{i}$, inflation $\boldsymbol{\pi}$ and taxes $\boldsymbol{\tau}$, the representative household picks sequences of consumption and asset holdings to maximize utility subject to her budget constraint as well as a standard no-Ponzi constraint. Show that optimal consumption behavior induces the following log-linearized Euler equation relationship:

$$\widehat{c}_t = \widehat{c}_{t+1} - \frac{1}{\gamma}\left(\widehat{i}_t - \widehat{\pi}_{t+1}\right). \tag{8}$$

b) Let $\gamma = 1$. Begin by computing and plotting impulse response functions to the monetary policy shock in the RA-PIH economy using standard first-order perturbation (state-space) techniques, e.g. using *dynare* or *gensys*. Does the assumed fiscal rule matter for equilibrium dynamics? What would happen to output, inflation, and nominal interest rates if instead of adjusting taxes to ensure that $\widehat{b}_t = 0$ the fiscal authority would instead adjust taxes so that

$$\widehat{b}_t = \rho_b \widehat{b}_{t-1} + (1 + \bar{r})(\widehat{i}_{t-1} - \widehat{\pi}_t), \tag{9}$$

where $\rho_b \in (0, 1)$? Discuss.

c) Now code up the same economy using sequence-space techniques. This will in particular require you to arrive at derivative matrices $\mathcal{C}_\bullet$ for the household's aggregate consumption function, satisfying

$$\widehat{\boldsymbol{c}} = \mathcal{C}_y \widehat{\boldsymbol{y}} + \mathcal{C}_i \widehat{\boldsymbol{i}} + \mathcal{C}_\pi \widehat{\boldsymbol{\pi}} + \mathcal{C}_\tau \widehat{\boldsymbol{\tau}}. \tag{10}$$

Begin by providing a closed-form expression for the entries of the matrix $\mathcal{C}_y$. What's the intuition for your expression? Then numerically construct all of $\{\mathcal{C}_y, \mathcal{C}_i, \mathcal{C}_\pi, \mathcal{C}_\tau\}$ using

your favorite programming language. With those matrices in hand you can use sequence-space techniques to compute and plot transition paths (= impulse responses) following the monetary policy shock. Those impulse responses should be exactly the same as what you found using state-space techniques in item b). Finally, decompose the consumption impulse response into the *direct* effects of monetary policy—the response of consumption to the change in real rates, i.e. $\mathcal{C}_i\widehat{\boldsymbol{i}} + \mathcal{C}_\pi\widehat{\boldsymbol{\pi}}$—and the *indirect* effects—the response to changes in income, i.e. $\mathcal{C}_y\widehat{\boldsymbol{y}} + \mathcal{C}_\tau\widehat{\boldsymbol{\tau}}$.[2] Discuss your results.

d) We now complicate the household block, changing preferences to the following "bond-in-utility" specification:

$$\sum_{t=0}^{\infty}\beta^t\left\{\frac{c_t^{1-\gamma}-1}{1-\gamma}+\alpha\frac{b_{t-1}^{1-\eta}-1}{1-\eta}\right\}. \tag{11}$$

In words, households now derive utility both from consumption and *directly* from their wealth holdings. We wish to understand how this change in preferences affects the transmission of the monetary policy shock. Throughout we will set $\gamma = \eta = 1$ and $\alpha = 0.1$.

(i) Find the value of β that is consistent with the steady state of the model. That is, find β such that, if $y_t = \bar{y}$, $\tau_t = \bar{\tau}$, $i_t = \bar{i}$ and $\pi_t = \bar{\pi}$ for all t, then the representative consumer finds it optimal to set $c_t = \bar{c}$ and $b_t = \bar{b}$ for all t. What can you say about $\beta(1+\bar{r})$? Is it still equal to 1? What's the intuition?

(ii) Derive the log-linearized Euler equation for this bond-in-utility representative household. How has the Euler equation changed relative to (8)?

(iii) Compute the consumption derivative matrices $\{\mathcal{C}_y, \mathcal{C}_i, \mathcal{C}_\pi, \mathcal{C}_\tau\}$. How does $\mathcal{C}_y$ compare with its analogue from the baseline permanent-income RA model studied above? Focus your discussion on the first column of $\mathcal{C}_y$.

(iv) Using sequence-space methods, compute and plot impulse responses to the monetary shock. How have they changed relative to the standard model considered above? What about the direct-indirect decomposition?

(v) Finally consider again the case where the fiscal authority instead follows a fiscal rule like (9). For concreteness set $\rho_b = 0.7$. Do the overall impulse responses to a monetary policy shock now depend on the fiscal rule? Why? And what happens as you increase ρ_b? Provide an economic intuition.

Question 2

As in Question 1 we consider a variant of the textbook New Keynesian model, this time throughout with Ricardian, permanent-income households. The point of this question will

[2] For further background on this you may consider reading the "HANK" paper of Kaplan et al. (2018).

be to get you to work with the policy counterfactual identification results in Lecture Note 4.

The familiar three-equation core of the model is

$$y_t = \mathbb{E}_t[y_{t+1}] - \frac{1}{\gamma}(i_t - \mathbb{E}_t[\pi_{t+1}]), \tag{12}$$

$$\pi_t = \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}] + e_t^s, \tag{13}$$

$$i_t = \rho_i i_{t-1} + (1-\rho_i)\phi_\pi \pi_t, \tag{14}$$

where all variables are in log deviations from steady state. Note that the sole shock to the system is the cost-push wedge e_t^s, which is assumed to follow an ARMA(1,1) process,

$$e_t^s = \rho_s e_{t-1}^s + \varepsilon_t^s + \theta_s \varepsilon_{t-1}^s, \quad \varepsilon_t^s \sim N(0, \sigma_s^2), \tag{15}$$

with ε_t^s denoting the cost-push innovation. The final part of the model is the government budget. Combining the budget constraint and a standard tax rule as in Leeper (1991), the fiscal block can be written as

$$b_t = \frac{1-\tau_d}{\beta} b_{t-1} + \frac{\bar{b}}{\bar{y}}(i_{t-1} - \pi_t). \tag{16}$$

Throughout this problem we will consider the following parameterization of the private sector block: $\beta = 0.99^{1/4}$, $\gamma = 1$, $\kappa = 0.02$, $\bar{b} = \bar{y} = 1$. For policy, we will begin by assuming $\phi_\pi = 1.7$, $\rho_i = 0.8$, and $\tau_d = 0.4$. We will consider several alternatives later. Finally, let us fix the parameters of the cost-push process to $\rho_s = 0.7$, $\theta_s = 0.4$, $\sigma_s = 1$.

a) Using sequence-space techniques, compute the impulse responses of output, inflation, and interest rates to the following two sets of shocks:

   (i) the cost-push shock ε_t^s,

   (ii) contemporaneous and news shocks to the monetary policy rule (14).

   Plot your results. For (ii), it suffices to plot impulse responses for shocks at some selected horizons, say $\{0, 2, 5, 10\}$.

b) Using the objects computed in a), evaluate the propagation of the cost-push shock ε_t^s under the counterfactual monetary policy rule $\phi_\pi = 3$, $\rho_i = 0$ together with an unchanged fiscal response. Plot the impulse responses of output, inflation, and interest rates, together with the sequence of shocks to the original rule (14) that implements the new rule. Verify your results against a direct solution of the model with the new, counterfactual monetary policy rule.

c) We now wish to evaluate the propagation of the cost-push shock ε_t^s under an interest rate peg, i.e., $\phi_\pi = \rho_i = 0$. On its own, such a counterfactual rule induces indeterminacy. We want to evaluate impulse responses under two assumptions on equilibrium selection.

(i) *Monetary dominance.* To compute outcomes under monetary dominance, consider an interest rate peg for a large but finite number of time periods, and then change the monetary rule back to the original Taylor rule (with $\phi_\pi = 1.7$, $\rho_i = 0.8$). Do your conclusions depend on the exact active monetary rule you assume at the large future date?

(ii) *Fiscal dominance.* Now suppose again that the peg is in place forever, but that also we set $\tau_d = 0$. First of all show that stability of government debt then requires that

$$\sum_{t=0}^{\infty} \beta^t \left(i_{t-1} - \pi_t\right) = 0, \tag{17}$$

with the convention that $i_{-1} = 0$. Now use this additional requirement to select among the many bounded equilibria consistent with the interest rate peg.

Question 3

Consider a stationary SVAR model

$$A(L)Y_t = H\varepsilon_t, \quad A(L) = I_n - \textstyle\sum_{\ell=1}^{\infty} A_\ell L^\ell, \quad \varepsilon_t \sim WN(0, I_n) \tag{18}$$

where A_ℓ is $n \times n$ for all ℓ, and H is $n \times n$ and non-singular.

Let $Y_{j,t}$ and $\varepsilon_{j,t}$ denote the j-th elements of Y_t and ε_t, respectively. Denote the (i,j) element of H by H_{ij}.

a) Assume that $n = 3$ and H is lower triangular (recursive identification). Show that

$$\varepsilon_{2,t} = \frac{1}{H_{22}} \Big\{ Y_{2,t} - \mathbb{E}^*(Y_{2,t} \mid Y_{1,t}, \{Y_\tau\}_{-\infty < \tau \leq t-1}) \Big\}.$$

b) Now drop the assumptions in a), and assume instead that we have access to a scalar variable Z_t that is jointly covariance stationary with ε_t and satisfies

$$\mathbb{E}(Z_t \varepsilon_{1,t}) \neq 0, \quad \mathbb{E}(Z_t \varepsilon_{j,t}) = 0 \text{ for all } j \geq 2.$$

Show that the *absolute* impulse responses

$$\Theta_{i,1,\ell} \equiv \mathbb{E}^*(Y_{i,t+\ell} \mid \varepsilon_{1,t} = 1) - \mathbb{E}^*(Y_{i,t+\ell} \mid \varepsilon_{1,t} = 0), \quad i = 1, \ldots, n, \ \ell \geq 0,$$

are identified. Show also that the forecast variance decompositions

$$FVD_{i,1,\ell} \equiv 1 - \frac{\text{Var}^*(Y_{i,t+\ell} \mid \{\varepsilon_{1,\tau}\}_{t<\tau\leq t+\ell}, \{\varepsilon_\tau\}_{-\infty<\tau\leq t})}{\text{Var}^*(Y_{i,t+\ell} \mid \{\varepsilon_\tau\}_{-\infty<\tau\leq t})}, \quad i = 1, \ldots, n,\ \ell \geq 1,$$

are identified. Give explicit formulas for these objects that could be implemented in the data $\{Y_t, Z_t\}$.

Question 4

The data set `q4_data.csv` contains data from the paper "Monetary Policy Surprises, Credit Costs, and Economic Activity" by Gertler & Karadi (2015, *American Economic Journal: Macroeconomics* 7(1), 44–76). The data is monthly and spans the time period 1990:1–2012:6. The time series are as follows:

`ipgr`	growth rate of industrial production, in annualized percentage points
`infl`	CPI inflation, in annualized percentage points
`gs1`	1-year government bond rate, in annualized percentage points
`ebp`	Excess Bond Premium (EBP), in annualized percentage points
`ff4_tc`	High-frequency changes in Federal Funds Futures prices

The variable `ebp` is a corporate bond spread that has been cleansed of default risk. The variable `ff4_tc` equals the monthly sum of high-frequency changes in three-month-ahead Federal Funds Futures prices measured in a short time window around Federal Open Market Committee (FOMC) monetary policy announcements (see Gertler & Karadi for details on the construction). `ff4_tc` equals 0 in months for which there is no FOMC meeting.

We seek to estimate the dynamic influence of monetary policy shocks on the EBP. We will use `ff4_tc` as a proxy/IV for the monetary shock. Collect the remaining four variables in the vector Y_t. In the following, include 4 lags of the data and an intercept in all VARs and regressions.

Assume first that the monetary policy shock is invertible with respect to $\{Y_t\}$, so that we can apply SVAR-IV.

a) Estimate the impulse response function (IRF) of the EBP to a monetary policy shock, for horizons $\ell = 0, 1, \ldots, 36$ months. Normalize the IRF such that it corresponds to a monetary shock that raises the 1-year government bond rate by 100 basis points on impact. Plot your results.

b) Estimate the Forecast Variance Decomposition (FVD) of the EBP with respect to the monetary policy shock, at forecast horizons $\ell = 1, 2, \ldots, 36$ months. Plot your results.

Now we drop the invertibility assumption.

c) Estimate the IRF in a) using an internal IV approach, implemented by ordering the IV first in a recursive VAR that also includes Y_t. Plot your results.

d) Test the null hypothesis that the monetary policy shock is invertible, using the VAR specification in c). You may assume i.i.d. innovations.

e) Repeat c), but implement the internal IV strategy using LP-IV-style 2SLS regressions. Remember to control for lags of Y_t and of the IV.

Question 5

In this exercise you will combine insights from Question 1 and Question 4 to interpret aggregate inflation dynamics through the lens of simple Phillips curve relationships.

a) Return to the standard Phillips curve considered in Question 1:

$$\widehat{\pi}_t = \kappa \widehat{y}_t + \frac{1}{1+\bar{r}} \widehat{\pi}_{t+1} \tag{19}$$

Re-arranging (19), it can be shown that inflation and output paths $\{\widehat{\boldsymbol{\pi}}, \widehat{\boldsymbol{y}}\}$ in any perfect-foresight transition path equilibrium of Question 1 satisfy the sequence-space relationship

$$\widehat{\boldsymbol{\pi}} = \Pi_y \times \widehat{\boldsymbol{y}} \tag{20}$$

for some matrix $\Pi_y(\kappa, \bar{r})$ that depends only on κ and $\bar{r}$. Provide an explicit closed-form expression for that matrix.

b) Let $\widehat{\boldsymbol{y}}_m$ denote the impulse response of industrial production growth to a monetary policy shock. For the remainder of this exercise you should fix this impulse response at the internal IV recursive VAR OLS point estimate from part c) of Question 4. Combine $\widehat{\boldsymbol{y}}_m$ with $\Pi_y(\kappa, \bar{r})$ for several "reasonable" values of $(\kappa, \bar{r})$ to construct $\widehat{\boldsymbol{\pi}}_m^{NKPC}$ through the NKPC relationship (20). How do the constructed $\widehat{\boldsymbol{\pi}}_m^{NKPC}$ compare with the actual empirically estimated shock response $\widehat{\boldsymbol{\pi}}_m$? Can you find any combination of parameter values that provides an adequate fit? What is your conclusion about the ability of simple forward-looking Phillips curve relationships like the textbook relation (19) to adequately describe joint aggregate (conditional) inflation and output dynamics?

Question 6

Consider a standard growth model with a representative household that maximizes expected log utility. The consumption-savings problem is

$$\max_{\{c_t,k_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \log(c_t)\right] \tag{21}$$

subject to

$$c_t + k_t = (1-\tau_t)k_{t-1}^{\alpha} + t_t \tag{22}$$

where: c_t denotes consumption; k_t denotes the capital stock; t_t denotes lump-sum transfers; and τ_t is the income tax rate. Transfers satisfy $t_t = \tau_t k_{t-1}^{\alpha}$ (balanced government budget). We will also write $y_t \equiv k_{t-1}^{\alpha}$ as total output. Note that we are assuming full depreciation of capital between periods; furthermore, we as usual assume that $\alpha, \beta \in (0,1)$. The income tax rate τ_t follows some (as-of-yet unspecified) stochastic process.

a) Show that the equilibrium conditions characterizing a solution to this growth model are

$$\frac{1}{c_t} = \alpha\beta\mathbb{E}_t\left[(1-\tau_{t+1})\frac{1}{c_{t+1}}\frac{y_{t+1}}{k_t}\right] \tag{23}$$

$$c_t + k_t = y_t = k_{t-1}^{\alpha} \tag{24}$$

b) We will now log-linearize this system around a deterministic steady state in which $\tau_t = \bar{\tau} \in (0,1)$ for all t. Eliminate y_t from (23) and log-linearize (23) - (24) in (c_t, τ_t, k_t).[3] Then combine the two linearized relations to arrive at the following second-order difference equation for the aggregate capital stock:

$$\mathbb{E}_t\left[\widehat{k}_{t+1}\right] - (\theta^{-1}+\alpha)\widehat{k}_t + \alpha\theta^{-1}\widehat{k}_{t-1} = \left\{\theta^{-1}(1-\theta)\frac{\bar{\tau}}{1-\bar{\tau}}\right\}\mathbb{E}_t\left[\widehat{\tau}_{t+1}\right] \tag{25}$$

where $\theta \equiv \alpha\beta(1-\bar{\tau})$. Note that under our assumptions $\theta \in (0,1)$.

c) Verify that

$$\widehat{k}_t = \alpha\widehat{k}_{t-1} - \underbrace{(1-\theta)\left(\frac{\bar{\tau}}{1-\bar{\tau}}\right)}_{\equiv\kappa}\sum_{i=0}^{\infty}\theta^i\mathbb{E}_t\left[\widehat{\tau}_{t+i+1}\right] \tag{26}$$

is *a* valid solution to the expectational difference equation (25).[4]

[3] To be clear, I want the expansion to be in log-deviations from steady state for all variables (c_t, τ_t, k_t).

[4] Evaluating the characteristic polynomial of (25) it can be shown that it is in fact the unique saddle path solution; you do not need to do so, however.

We will now specialize the tax process. We assume that taxes are i.i.d., but that they are announced two periods in advance. That is, we have

$$\widehat{\tau}_t = \varepsilon_{\tau,t-2} \tag{27}$$

and so

$$\widehat{k}_t = \alpha \widehat{k}_{t-1} - \kappa \left\{ \varepsilon_{\tau,t-1} + \theta \varepsilon_{\tau,t} \right\} \tag{28}$$

Intuitively, this set-up reflects legislative delays—tax changes come into effect a little bit after they have been announced.

d) Since $\theta \in (0,1)$ (28) implies that recent tax news $\varepsilon_{\tau,t}$ have a smaller effect on capital accumulation today than older news $\varepsilon_{\tau,t-1}$. What's the economic intuition?

e) Is this process invertible? That is, do we have that

$$\varepsilon_{\tau,t} \in \text{span}\left(\{\widehat{k}_{t-\ell}\}_{\ell=0}^{\infty} \right) \tag{29}$$

Discuss.

f) Is the process recoverable? That is, do we have that

$$\varepsilon_{\tau,t} \in \text{span}\left(\{\widehat{k}_{t-\ell}\}_{\ell=-\infty}^{\infty} \right) \tag{30}$$

Discuss.